

CHAPTER 5: CONCLUSION

In this study, we conduct two research tasks. One is to examine the impact of the operation of Green Line LRT on the AADT of its adjacent roads, and the other is to estimate PNR location choice model in the Twin Cities area.

In the first task, we applied a quasi-experiment (before-after, treatment-control) research design to examine the effects of the Green Line LRT on vehicular travel demand on the roads in its vicinity. We developed two difference-in-difference models with a negative binomial link function. The first model emphasized annual average daily traffic (AADT) before and after the opening of the Green Line. The second one examined AADT changes over three time periods. To identify the independent effects of the Green Line on AADT, both models controlled for confounding variables, including road classification, land use, and transit supply. This task offered critical insights into the effect of rail transit on road traffic.

- All else being equal, the Green Line reduced AADT of adjacent road segments by 18%. This finding emphasized the critical role of rail transit in travel demand management.
- While the Green Line reduced road traffic in the first two years of operation, the effect became smaller in the next two years. Specifically, the effect decreased from 22% to 16%. The decreasing effect was likely attributable to land-use development induced by the Green Line and travel demand that switched from other routes and modes. Given that more development will occur in the corridor, we expect the effect of the Green Line on road traffic to continue to decrease, a hypothesis to be tested.
- It is crucial to control for confounding factors such as road classification, land use, and transit supply. Otherwise, we are likely to substantially bias the estimates of rail transit effect.